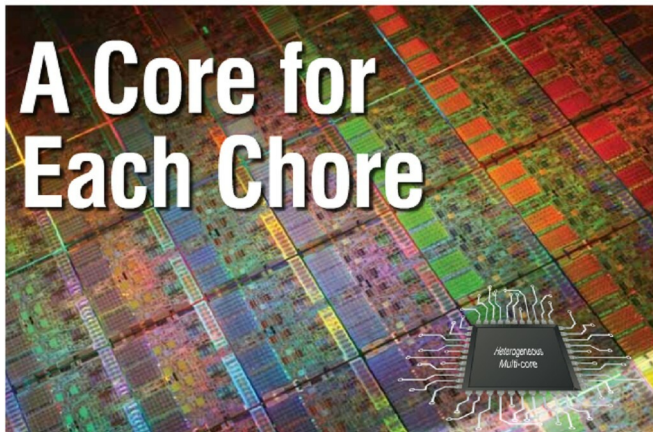




Dual- and quad-core processors will soon be old hat, with industry majors working towards six-core desktop CPUs and eight- and 12-core server processors, not to forget the prototypes with dozens of cores. Ongoing research also promises highly scalable and customisable heterogeneous multi-core products in the future, complemented with the right parallel programming tools to take advantage of these capabilities.

*H*ave you ever wished you had four hands so you could work faster? A similar sentiment amongst computer engineers was what led to the development of multi-core processors more than a decade ago.

A multi-core processor is essentially a chip that has more than one processor (core) attached to it. By engaging these multiple cores using the right software, a computer can do more in less time.

There is a great demand for multi-core processors across various application domains ranging from general-purpose and embedded computing to networking, digital signal processing (DSP) and graphics. Over the past few years, multi-cores have become the norm, with dual- and quad-core processors being taken for granted even in the desktop and mobile computing spaces.

Of course, we all know that in the computing

world, users are never satisfied. However fast their computers are, they want more speed.

When they find four cores mundane, they will have six-core processors, but soon even that will not be enough. Every need will be met by the industry, with more research, more innovations and more products. It is an unending cycle, and no technology is permanent.

Let us therefore stop at this moment in time and capture the current state of affairs in multi-core processors. The situation here is quite similar to the Cola Wars, with Intel and AMD trying to outdo each other with more cores.

A six-mare chariot to pull your computer

Six- and even eight-core processors, per se, are not news any more, with server-focused products such as Intel's Xeon and AMD's Opteron being quite popular. However, it is interesting to note that Intel recently demonstrated a six-core